Environment benchmark: repeated runs (mean ± standard deviation)

Each value is the mean ± the standard deviation, over the runs, of a per-run figure: the percent of all questions in the set with that voted label, the total cost of the run, and the mean answer time. Configurations 1–8: masked correctness on set A (141 questions with a value hidden from the case), factual consistency on set B (272 questions the case never answers). Configuration 9: both on set C (272 questions the case answers).

Environment configuration	Runs	Masked correctness				Factual consistency			Environment cost and time	
		same category (incl. exact)†	different category	not comparable	no label*	consistent	inconsistent	no label*	Total cost per run (USD)‡	Average answer time (s)
1. Gemini 3 Flash knowledge, no search (gemini-3-flash-preview)	10	84.4 ± 1.9%	13.8 ± 1.1%	1.1 ± 1.0%	0.7 ± 1.5%	97.3 ± 1.5%	0.1 ± 0.2%	2.6 ± 1.5%	$1.067 ± 0.117	70.3 ± 7.6
2. Current without changes 1–4 (Europe PMC + LitSense)	10	86.0 ± 1.6%	12.8 ± 1.6%	0.9 ± 0.9%	0.4 ± 0.5%	97.0 ± 1.2%	0.0 ± 0.1%	3.0 ± 1.2%	$0.973 ± 0.131	122.7 ± 13.3
3. Current with changes 1–4 (Europe PMC + LitSense)	10	85.0 ± 1.6%	14.0 ± 1.5%	0.7 ± 0.7%	0.4 ± 0.4%	97.1 ± 1.0%	0.4 ± 0.3%	2.5 ± 0.8%	$1.032 ± 0.176	106.8 ± 5.1
4. OpenRouter search: Exa instant, DeepSeek v4 Flash	10	85.8 ± 2.0%	13.2 ± 2.1%	0.6 ± 0.4%	0.4 ± 0.5%	97.2 ± 0.9%	0.1 ± 0.3%	2.6 ± 0.9%	$3.802 ± 0.172	78.9 ± 10.9
5. OpenRouter search: Parallel basic, DeepSeek v4 Flash	10	83.3 ± 2.4%	15.2 ± 2.5%	1.1 ± 1.0%	0.4 ± 0.4%	97.2 ± 1.0%	0.2 ± 0.3%	2.6 ± 0.8%	$3.062 ± 0.138	83.8 ± 5.5
6. OpenRouter search: Perplexity, DeepSeek v4 Flash	10	83.5 ± 2.2%	15.5 ± 2.3%	0.8 ± 0.5%	0.3 ± 0.5%	97.5 ± 0.7%	0.1 ± 0.2%	2.4 ± 0.7%	$3.220 ± 0.147	93.4 ± 8.3
7. OpenRouter search: Google (native), Gemini 3.1 Flash Lite	1	83.0%	16.3%	0.7%	0.0%	96.7%	0.0%	3.3%	$21.086	82.4
8. OpenRouter search: OpenAI (native), GPT-6-luna	1	85.1%	14.9%	0.0%	0.0%	95.6%	0.0%	4.4%	$10.800	79.9
9. Case information only, no retrieval (DeepSeek v4 Flash 0731)	10	99.2 ± 0.4%	0.4 ± 0.2%	0.3 ± 0.3%	0.1 ± 0.2%	99.7 ± 0.3%	0.3 ± 0.2%	0.1 ± 0.2%	$0.053 ± 0.005	7.8 ± 0.8

- Every run answers the same questions; runs 2 and later search anew (no cached search results). The standard deviation is the sample standard deviation (n – 1) across runs.
- Configurations 7 and 8 (native Google and OpenAI search) were run once, so they have no standard deviation, and their cost is run 1's, at the earlier (lower) DeepSeek price.
- Judge panel: google/gemini-3.5-flash-lite, qwen/qwen3.8-flash, openai/gpt-6-luna; each answer's label is the majority vote (with no majority, the first judge decides).
- † Same category (incl. exact): answers at least in the true value's category (exact + same category). With different category, not comparable, and no label it sums to 100%.
- * No label: medsim gave no answer (almost all are questions its query builder refuses), or the run failed; it is not a judge label.
- Total cost per run: every environment component for all questions of the configuration (medsim Stages A–C, search fees, the search or knowledge model's tokens), from OpenRouter's usage.cost; judging is not included. Average answer time: mean wall time per question.
- ‡ Cost is the mean ± SD over runs 2 and later only: after run 1 (2026-09-29) OpenRouter raised deepseek/deepseek-v4-flash-0731's output price from $0.32 to $1.28 per million tokens, so run 1's cost is at another price level. Run 1 cost: 1: $0.66; 2: $0.44; 3: $0.42; 4: $3.26; 5: $2.52; 6: $2.47; 9: $0.03 (by row number). Percentages and time use every run.
- Run 1 reached the best-guess fallback through a replay of its declined answers on the same documents; in runs 2 and later it is built into medsim.
- 12 of the judge calls in runs 2 and later failed on every attempt (qwen/qwen3.8-flash rate limits); those answers were voted on by the other two judges.